The Breast 82 (2025) 104483

Contents lists available at ScienceDirect

The Breast

journal homepage: www.journals.elsevier.com/the-breast

Options for postoperative radiation therapy in patients with de novo metastatic breast cancer

Chaofan Li [a,1], Yusheng Wang [b,1], Biyun Fang [a], Mengjie Liu [a], Shiyu Sun [a], Jingkun Qu [a], Shuqun Zhang [a,*], Chong Du [a,**]

[a] The Comprehensive Breast Care Center, The Second Affiliated Hospital of Xi'an Jiaotong University, 157 West Fifth Street, Xi'an, Shaanxi, PR China
[b] Department of Otolaryngology, The Second Affiliated Hospital of Xi'an Jiaotong University, 157 West Fifth Street, Xi'an, Shaanxi, PR China

ARTICLE INFO

Keywords:
de novo metastatic breast cancer (dnMBC)
Postoperative radiation therapy
Random survival forest (RSF)
Shiny
Propensity score matching (PSM)

ABSTRACT

Background: Although meta-analyses have demonstrated survival benefits associated with primary tumor resection in MBC, guidelines lack consensus on the survival benefit of postoperative radiation therapy (RT).

Methods: In this study, we included 1392 patients with de novo metastatic breast cancer (dnMBC) by integrating data from the SEER database (2010–2019) to systematically assess the efficacy of postoperative RT and develop a machine learning-driven prognostic tool. The primary endpoint was overall survival (OS).

Results: Propensity score matching (PSM) results showed that postoperative RT significantly improved OS (HR = 0.573, 95 % CI = 0.475–0.693), but this survival gain showed great heterogeneity among different subgroups. It is found that patients with HR-/HER2-or HR+/HER2-subtypes gained significant OS benefit from ($p < 0.001$) postoperative RT, whereas patients with HER2+ subtype did not gain any survival benefit since the effect of targeted therapy overshadowed the postoperative RT. Further risk stratification by the random survival forest (RSF) model revealed that high-risk patients with T4/N3 stage, high tumor grade and poor response to chemotherapy had significantly prolonged OS after receiving RT ($p < 0.001$), while low-risk patients showed no additional benefit. The model had excellent predictive efficacy (training set C-index = 0.741, validation set C-index = 0.720) with key predictors including HER2 status, chemotherapy response and tumor grade. The research team developed an interactive web application (https://lee2287171854.shinyapps.io/RSFshiny/) based on this model, which can generate individualized survival risk scores in real-time to guide clinical decision-making.

Conclusion: This study is the first to propose a risk stratification strategy for postoperative RT in dnMBC, and innovatively integrates machine learning and clinical tools to provide a new paradigm for optimizing precision therapy.

1. Introduction

De novo metastatic breast cancer (dnMBC), defined by the presence of distant metastases at initial diagnosis, represents a distinct clinical entity accounting for 5–10 % of newly diagnosed breast cancer cases [1,2]. Despite its inherently advanced stage, emerging evidence suggests that dnMBC patients exhibit a more favorable prognosis compared to those with recurrent metastatic disease, with a 5-year disease-specific survival (DSS) rate of 44 % versus 21 % and median overall survival (OS) of 39.2 versus 27.2 months [3,4]. However, current clinical trials and prognostic frameworks for metastatic breast cancer (MBC) often conflate dnMBC with recurrent MBC, potentially obscuring critical survival dynamics unique to dnMBC and hindering the development of tailored therapeutic strategies [5,6]. This underscores an unmet need for prognostic tools specifically designed for dnMBC to guide precision medicine.

While systemic therapy remains the cornerstone of dnMBC management, the role of locoregional interventions-particularly post-operative radiation therapy (RT)-remains controversial. Although meta-analyses have demonstrated survival benefits associated with primary

* Corresponding author.
** Corresponding author.
E-mail addresses: shuqun_zhang1971@163.com (S. Zhang), duchong@xjtu.edu.cn (C. Du).
[1] These authors have contributed equally to this work and share the first authorship.

https://doi.org/10.1016/j.breast.2025.104483
Received 26 February 2025; Received in revised form 19 April 2025; Accepted 22 April 2025
Available online 23 April 2025

Table 1
Baseline characteristics of dnMBC patients (treated with vs. without postoperative radiation therapy).

Characteristics		Postoperative radiation therapy				
		No		Yes		P Value
		N = 579	%	N = 813	%	
Age at diagnosis	<45	150	25.91 %	234	28.78 %	0.629
	45–54	165	28.50 %	229	28.17 %	
	55–64	161	27.81 %	207	25.46 %	
	≥65	103	17.79 %	143	17.59 %	
Race	White	411	70.98 %	580	71.34 %	0.124
	Black	110	19.00 %	129	15.87 %	
	Others	58	10.02 %	104	12.79 %	
T stage	T1	46	7.94 %	61	7.50 %	0.317
	T2	155	26.77 %	234	28.78 %	
	T3	133	22.97 %	199	24.48 %	
	T4	217	37.48 %	296	36.41 %	
	Unknown	28	4.84 %	23	2.83 %	
N stage	N0	79	13.64 %	75	9.23 %	0.002
	N1	248	42.83 %	350	43.05 %	
	N2	96	16.58 %	133	16.36 %	
	N3	147	25.39 %	253	31.12 %	
	Unknown	9	1.55 %	2	0.25 %	
Grade	I/II	199	34.37 %	282	34.69 %	0.991
	III/IV	310	53.54 %	434	53.38 %	
	Unknown	70	12.09 %	97	11.93 %	
Histological type	IDC	483	83.42 %	678	83.39 %	0.821
	ILC	27	4.66 %	33	4.06 %	
	Others	69	11.92 %	102	12.55 %	
Hormone receptor	Negative	216	37.31 %	268	32.96 %	0.105
	Positive	363	62.69 %	545	67.04 %	
HER2 status	Negative	352	60.79 %	497	61.13 %	0.943
	Positive	227	39.21 %	316	38.87 %	
Bone metastases	No	279	48.19 %	364	44.77 %	0.076
	Yes	292	50.43 %	445	54.74 %	
	Unknown	8	1.38 %	4	0.49 %	
Liver metastases	No	424	73.23 %	652	80.20 %	0.007
	Yes	143	24.70 %	152	18.70 %	
	Unknown	12	2.07 %	9	1.11 %	
Lung metastases	No	397	68.57 %	697	85.73 %	<0.001
	Yes	169	29.19 %	106	13.04 %	
	Unknown	13	2.25 %	10	1.23 %	
Brain metastases	No	559	96.55 %	778	95.69 %	0.003
	Yes	5	0.86 %	19	2.34 %	
	Unknown	15	2.59 %	6	0.74 %	
Response to chemotherapy	No response	111	19.17 %	125	15.38 %	0.024
	Partial response	322	55.61 %	434	53.38 %	
	Complete response	146	25.22 %	254	31.24 %	

tumor resection in MBC [7,8], recent studies evaluating combined locoregional therapy (LRT, encompassing surgery and RT) report conflicting results. For example, Reinhorn et al. found no significant survival benefit for LRT [9], while Gera et al. suggested that LRT may improve outcomes in selected metastatic breast cancer cohorts [10]. These discrepancies highlight the need for nuanced patient selection. However, current guidelines lack consensus on the utility of RT following surgery [11], and existing studies predominantly focus on surgical outcomes rather than postoperative RT [2,12–15]. Notably, 35–80 % of stage IV breast cancer patients undergo primary tumor resection, often accompanied by RT [2,12], yet the selection criteria for RT and its survival impact in dnMBC remain poorly defined. Compounding this uncertainty, no prognostic model currently exists to identify patients most likely to benefit from postoperative RT, reflecting a critical gap in personalized dnMBC care.

To address these challenges, we conducted a large-scale cohort study with dual objectives: to systematically evaluate the survival benefits of postoperative RT in dnMBC and delineate patient subgroups deriving maximal benefit; develop and validate a machine learning-driven prognostic tool–integrated into a user-friendly web application for real-time, individualized survival prediction.

2. Materials and methods

2.1. Data source and study design

This retrospective cohort study utilized data from the Surveillance, Epidemiology, and End Results (SEER) database (2010–2019). Patients diagnosed with de novo metastatic breast cancer (dnMBC) were included based on the following criteria: (1) female patients; (2) histologically confirmed invasive breast cancer with distant metastases at initial diagnosis; (3) receipt of systemic chemotherapy and surgery on primary; (4) response to chemotherapy was evaluated. Exclusion criteria included: (1) patients with non-primary breast cancer (recurrent metastatic disease); (2) incomplete follow-up or survival data; (3) any prior malignancies; (4) received preoperative radiation therapy.

2.2. Variables and definitions

Clinicopathological variables included age at diagnosis, race, tumor stage (T/N), histological type, grade, hormone receptor (HR) and HER2 status, metastatic sites (bone, liver, lung, brain), chemotherapy response (complete/partial/no response), and treatment details (postoperative radiation therapy). The main survival outcome is overall survival (OS), calculated from diagnosis to death or last follow-up.

Table 2
Univariate and multivariate Cox analysis of prognostic factors for the development of prognostic survival model.

Characteristics		Univariate Cox analysis			Multivariate Cox analysis		
		HR	95 %CI	P Value	HR	95 %CI	P Value
Age at diagnosis	<45	reference			reference		
	45–54	1.202	0.950–1.521	0.126	1.262	0.974–1.635	0.078
	55–64	1.323	1.047–1.673	*	1.180	0.907–1.534	0.217
	≥65	1.768	1.372–2.279	***	1.359	1.014–1.821	*
Race	White	reference			reference		
	Black	1.345	1.089–1.661	**	1.028	0.808–1.309	0.822
	Others	0.798	0.599–1.063	0.124	0.876	0.632–1.213	0.424
T stage	T1	reference			reference		
	T2	1.539	1.032–2.294	*	1.467	0.946–2.274	0.087
	T3	1.538	1.026–2.306	*	1.255	0.803–1.959	0.319
	T4	2.264	1.541–3.325	***	1.802	1.175–2.762	**
N stage	N0	reference			reference		
	N1	1.472	1.053–2.058	*	1.634	1.116–2.392	*
	N2	1.828	1.273–2.623	**	1.674	1.104–2.540	*
	N3	2.552	1.823–3.574	***	2.385	1.621–3.508	***
Grade	I/II	reference			reference		
	III/IV	1.554	1.289–1.873	***	1.542	1.251–1.901	***
Histological type	IDC	reference			reference		
	ILC	1.347	0.945–1.920	0.099	1.074	0.705–1.637	0.739
	Others	1.400	1.119–1.752	**	1.363	1.050–1.771	*
Hormone receptor	Negative	reference			reference		
	Positive	0.605	0.511–0.717	***	0.516	0.419–0.636	***
HER2 status	Negative	reference			reference		
	Positive	0.384	0.316–0.465	***	0.432	0.345–0.541	***
Bone metastases	No	reference					
	Yes	1.096	0.925–1.299	0.288	/	/	/
Liver metastases	No	reference					
	Yes	1.105	0.907–1.346	0.320	/	/	/
Lung metastases	No	reference			reference		
	Yes	1.506	1.243–1.824	***	0.951	0.752–1.202	0.673
Brain metastases	No	reference			reference		
	Yes	2.979	1.810–4.904	***	2.757	1.513–5.023	***
Postoperative radiation therapy	No	reference			reference		
	Yes	0.595	0.504–0.702	***	0.573	0.475–0.693	***
Response to chemotherapy	No response	reference			reference		
	Partial response	0.535	0.437–0.655	***	0.546	0.433–0.688	***
	Complete response	0.257	0.197–0.335	***	0.304	0.221–0.419	***

*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$.

2.3. Statistical analysis

Categorical variables were expressed as frequencies and percentages. The between-group comparison of categorical data was achieved using the $\chi 2$ test or Fisher's exact test. Univariate and multivariate Cox proportional hazards models identified prognostic factors for OS. Variables with $P < 0.05$ in univariate analysis were included in the multivariate model. Propensity score matching (PSM) adjusted for confounding variables (age, T/N stage, metastatic sites) was performed to compare OS between RT and non-RT groups. Survival rates were compared using Kaplan-Meier curves, and statistical significance was tested using the log-rank test.

2.4. Machine learning model development

Random Survival Forest (RSF) Model: Variables from multivariate Cox analysis were included. There was missing information for brain metastases, grade, T stage, and N stage, which was automatically omitted in the construction of the model as the RSF algorithm was unable to handle the missing values. After excluding patients with missing data (n = 224), 1168 patients were included in the model. The dataset was split into training (50 %) and test (50 %) sets. Hyperparameters (number of trees, node depth) were optimized via 10-fold cross-validation. Concordance index (C-index), Brier score, AUC value, and calibration curves assessed model accuracy. Decision curve analysis (DCA) evaluated clinical utility. Patients were classified into high- and low-risk groups using RSF-predicted mortality risk thresholds.

2.5. Web application development

The RSF model was deployed as a web application using R Shiny. Input fields included clinicopathological variables. The backend generated real-time survival risk scores. Five oncologists validated the interface for usability. All analyses were conducted in R studio. R codes are available upon request.

3. Results

3.1. Clinical characteristics and postoperative RT utilization

This study included 1392 dnMBC patients, whose clinicopathological characteristics stratified by post-surgery RT utilization are detailed in Table 1. Significant differences were observed between the RT and non-RT groups, particularly in N stage, metastatic patterns, and chemotherapy response. Patients with advanced N stage (N3) were more likely to receive post-surgery RT (31.12 % vs. 25.39 %, $p < 0.05$). Conversely, those with liver or lung metastases had a lower likelihood of receiving RT, while brain metastases were associated with a higher RT utilization rate ($p < 0.05$). Additionally, patients achieving a complete response to chemotherapy were more likely to undergo post-surgery RT (31.24 % vs. 25.22 %, $p < 0.05$), suggesting RT is selectively administered based on disease burden and treatment response.

3.2. Prognostic factors for overall survival (OS)

Univariate and multivariate Cox regression analyses were performed

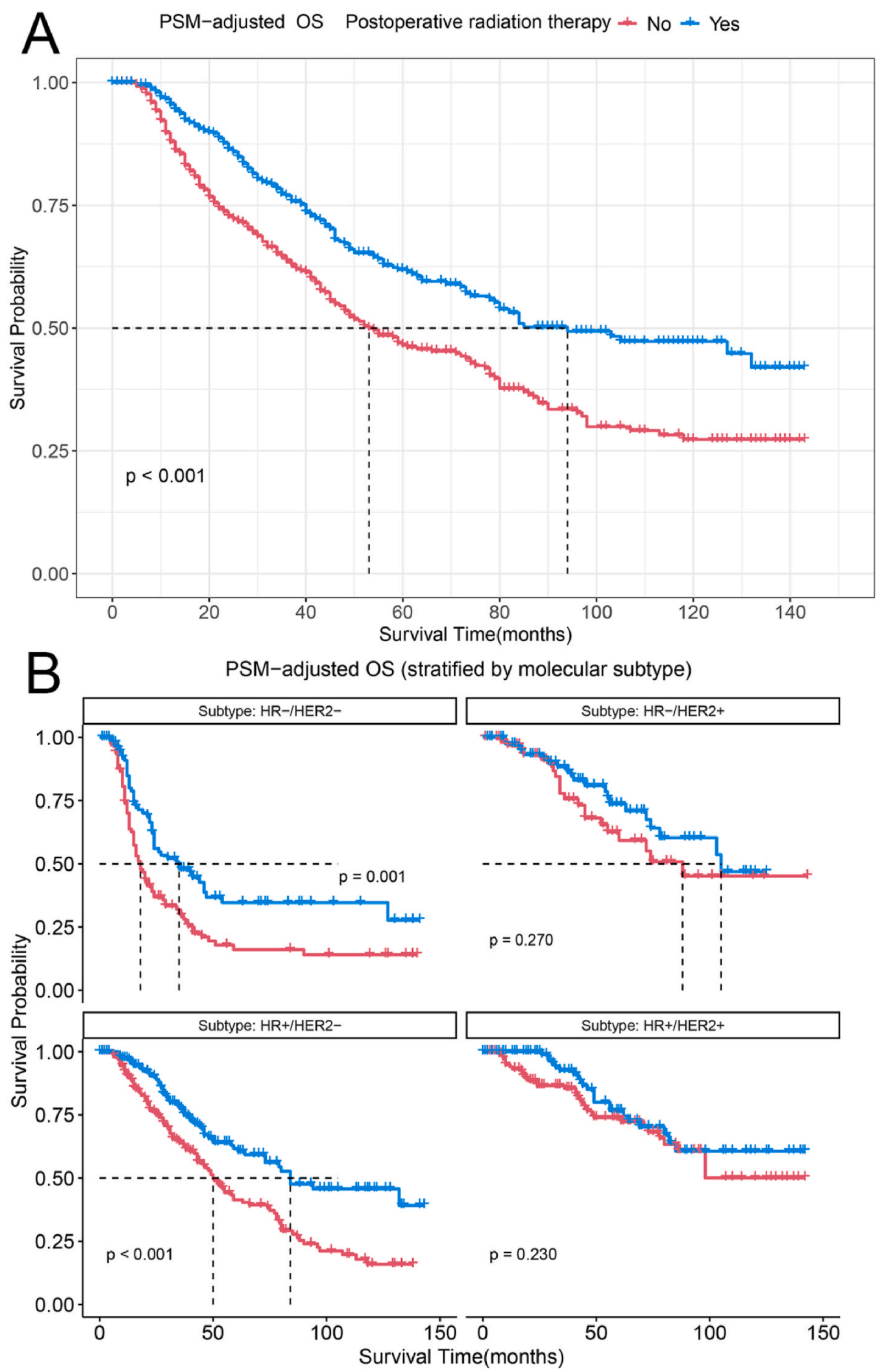

Fig. 1. Survival analysis after PSM adjustment for baseline characteristics (radiation therapy vs. no radiation therapy).
Kaplan-Meier (K–M) survival analysis:A. PSM-adjusted OS of patients; B. PSM-adjusted OS of patients (stratified by molecular subtype); PSM: Propensity score matching; OS: overall survival.

to identify prognostic factors for OS in dnMBC (Table 2). Univariate analysis revealed significant associations between OS and age, race, T/N stage, tumor grade, histological type, hormone receptor (HR) and HER2 status, metastatic sites, post-surgery RT, and chemotherapy response. Multivariate analysis further identified independent prognostic factors:

Poor OS predictors: Age ≥65 years (HR = 1.359, 95 % CI = 1.014–1.821), T4 stage (HR = 1.802, 95 % CI = 1.175–2.762), higher N stage (N3: HR = 2.385, 95 % CI = 1.621–3.509), grade III/IV (HR = 2.385, 95 % CI = 1.621–3.508), Other histological types (HR = 1.363, 95 % CI = 1.050–1.771), and brain metastases (HR = 2.757, 95 % CI = 1.513–5.023). Favorable OS predictors: HR+ (HR = 0.516, 95 % CI = 0.419–0.636), HER2+ (HR = 0.432, 95 % CI = 0.345–0.541), post-surgery RT (HR = 0.573, 95 % CI = 0.475–0.693), and better chemotherapy response (complete response: HR = 0.304, 95 % CI = 0.221–0.419).

3.3. *Stratified analysis of post-surgery RT benefits*

To delineate the patient subgroups benefiting most from post-surgery RT, propensity score matching (PSM)-adjusted survival analysis was

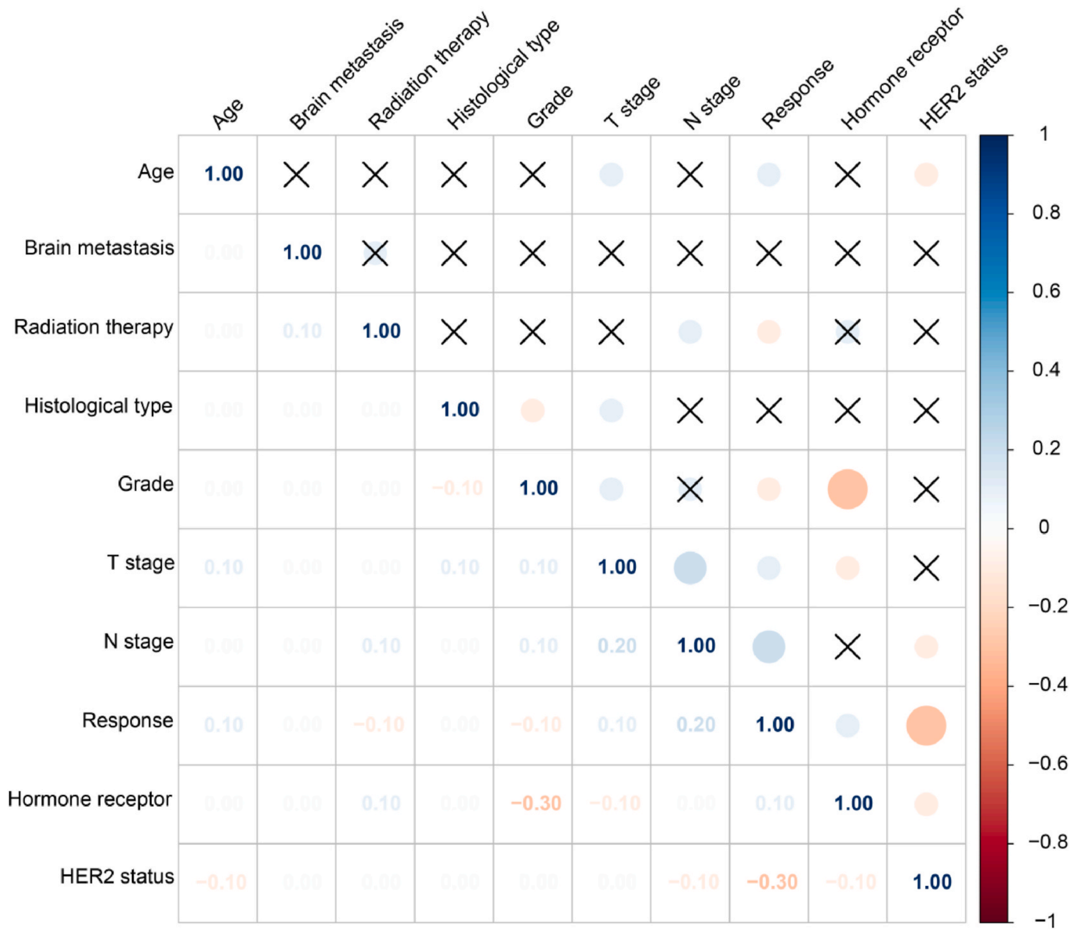

Fig. 2. The results of the correlation test between clinicopathological variables.
Spearman's correlation matrix illustrating associations between clinicopathological variables. Color intensity and circle size reflect correlation strength; "X" denote no statistical significance, and no "X" ($p < 0.05$ is considered statistically significant).

conducted. Patients receiving RT exhibited significantly improved OS compared to non-RT counterparts ($p < 0.001$, Fig. 1). However, this benefit was restricted to specific molecular subtype: HR-/HER2-and HR+/HER2-patients derived significant survival gains, whereas HER2+ patients showed no benefit. Further correlation analysis revealed strong associations between tumor grade and HR status, as well as between chemotherapy response and HER2 status (Fig. 2), underscoring the interplay between tumor biology and treatment efficacy.

3.4. Development and validation of the random survival forest (RSF) model

To enhance prognostic precision, we developed an RSF model incorporating the significant variables identified by multivariate Cox analysis. The model demonstrated robust predictive performance, with C-indices of 0.741 and 0.720 for the training and test sets, respectively (Fig. 3A and B). Fig. 3C demonstrates the RSF-predicted survival of the first five patients and Fig. 3D is the visualization of the first tree of RSF model. Fig. 3E is the error rate of the RSF model as a function of the number of trees. Key prognostic factors, including HER2 status, chemotherapy response, and tumor grade, were ranked as the top predictors (Fig. 3F). The model's superiority over traditional Cox analysis was further validated by: Brier scores (Fig. 4A), indicating lower prediction error. AUC values for 1-, 3-, and 5-year OS predictions on the test set were 0.8164, 0.7645, and 0.7165, respectively (Fig. 4C). Calibration curves in Fig. 4E show excellent agreement between predicted and observed survival probabilities. Decision curve analysis (Fig. 4B–D, F), showed that the model had a wide threshold probability range and well net benefit in predicting 1-year, 3-year, and 5-year OS.

3.5. Risk stratification and RT efficacy in high-risk patients

Patients were stratified into high- and low-risk groups based on RSF model predictions (Table 3). Kaplan-Meier analysis revealed significantly worse OS in the high-risk group ($p < 0.001$, Fig. 5A). Notably, post-surgery RT conferred a survival benefit exclusively in high-risk patients (Fig. 5B), suggesting that RT should be prioritized for this subgroup to maximize therapeutic impact.

3.6. Web-based application for clinical utility

To facilitate clinical adoption, we developed a user-friendly web application using the Shiny platform (Fig. 6). This tool enables clinicians to input patient clinicopathological features and obtain personalized survival predictions, thereby supporting tailored treatment strategies for dnMBC patients (https://lee2287171854.shinyapps.io/RSFshiny/).

4. Discussion

Our study provides pivotal insights into the management of dnMBC by addressing two unresolved questions: (1) the role of postoperative RT in a heterogeneous dnMBC population, and (2) the development of a precision prognostic tool to guide therapeutic decisions. The key findings and their implications are elaborated below.

It showed that postoperative RT is an independent prognostic factor for improved OS, particularly in HR-/HER2-and HR+/HER2-subtypes.

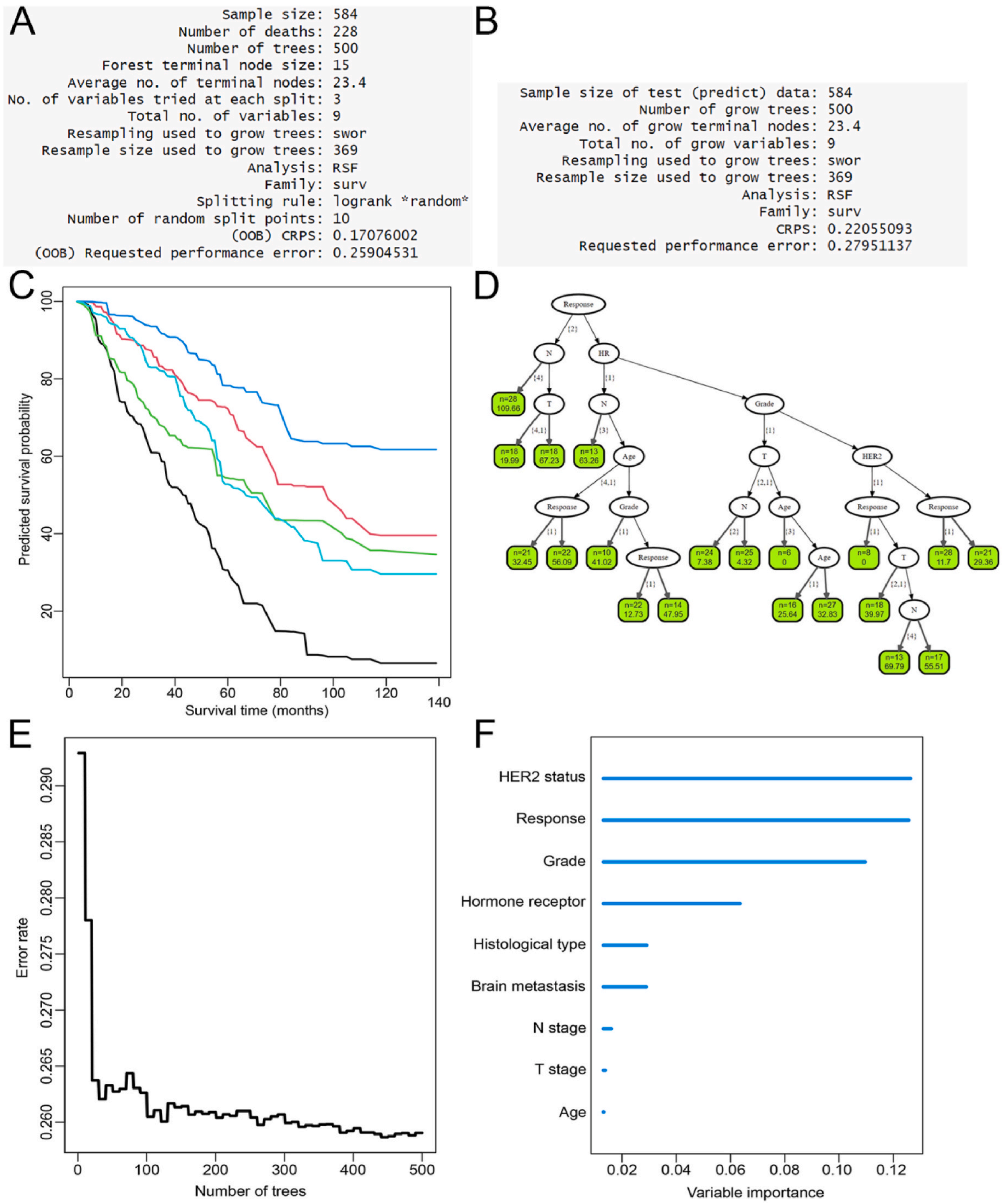

Fig. 3. Construction of random survival forest (RSF) model.
Model building process: A. performance of the model on the train set; B. performance of the model on the test set; C. the RSF-predicted survival of the first five patients; D. the visualization of the first tree of RSF model; E. the error rate of the RSF model as a function of the number of trees; F. feature importance ranking in RSF model.

This aligns with the hypothesis that locoregional control reduces systemic tumor burden by eliminating residual microscopic disease and suppressing metastatic seeding from the primary tumor [16,17]. For instance, the removal of primary tumors may disrupt the production of circulating tumor cells (CTCs) and cancer stem cells, both implicated in metastatic progression [17–20]. However, the absence of RT benefit in HER2+ patients likely reflects the dominant efficacy of HER2-targeted therapies (e.g., trastuzumab, pertuzumab), which may overshadow incremental gains from RT [21–23]. This dichotomy underscores the necessity of biomarker-driven stratification, as blanket application of RT risks overtreatment in HER2+ cohorts while withholding it may deprive high-risk HER2-patients of potential survival gains. What's more, emerging therapies, including CDK4/6 inhibitors, PARP inhibitors, and immune checkpoint blockers, are reshaping survival trajectories in dnMBC [24,25]. While our model accounts for HER2-targeted agents, it does not yet incorporate these novel therapies. Future work should explore the interplay between RT and emerging systemic therapies (e.g., CDK4/6 inhibitors) and integrate molecular biomarkers (e.g., PIK3CA

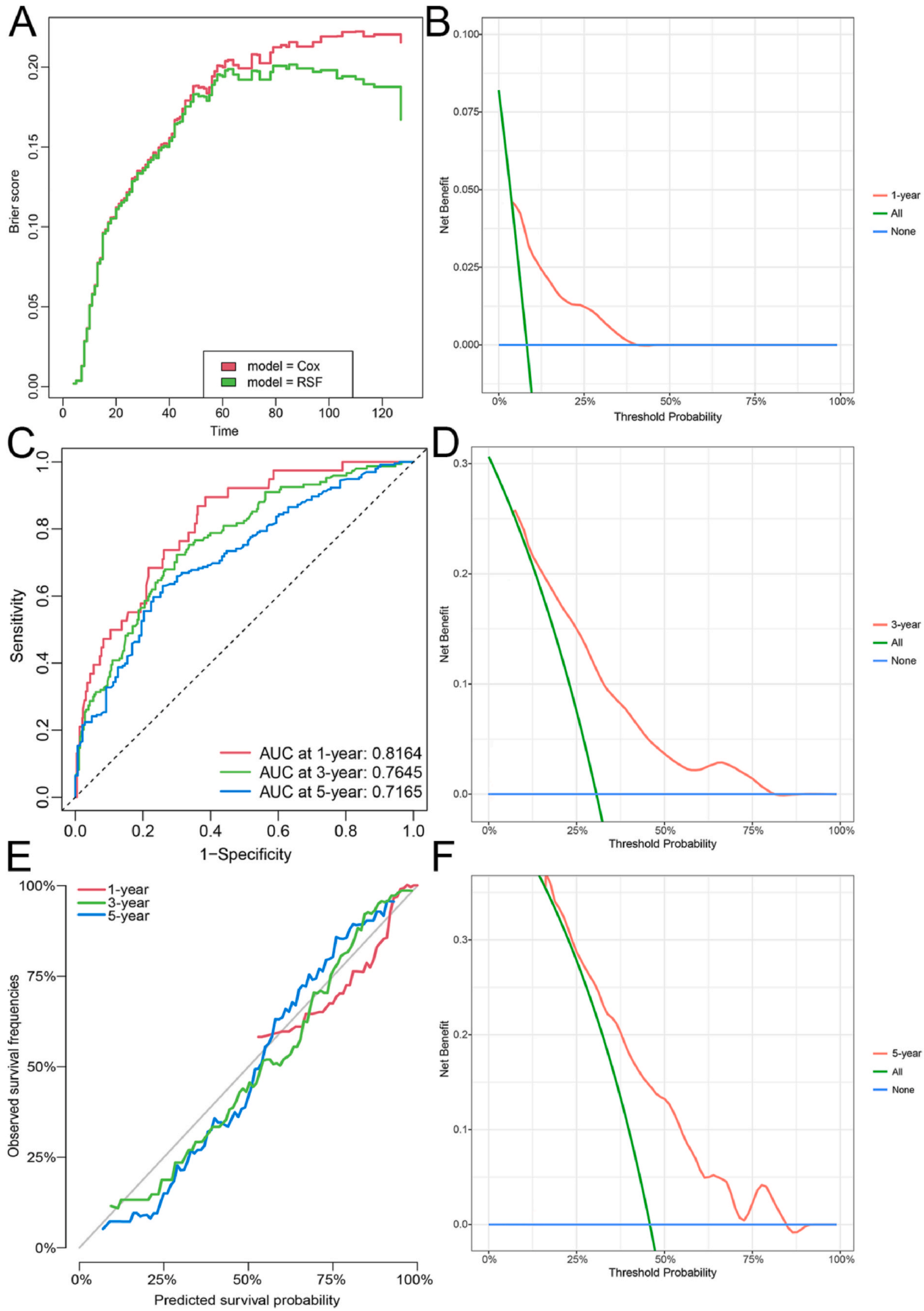

Fig. 4. Evaluation of random survival forest (RSF) model.
Model evaluation process: A. the model's superiority over traditional Cox analysis was further validated by: brier scores; B. decision curve analysis of 1-year survival on the test set; C. ROC curves and AUC values of 1-, 3, and 5-year survival on the test set; D. decision curve analysis of 3-year survival on the test set; E. calibration curves of 1-, 3, and 5-year survival on the test set; F. decision curve analysis of 5-year survival on the test set.

Table 3
Baseline characteristics of patients stratified by random survival forest.

Characteristics		RSF-adjusted risk stratification				
		High		Low		P Value
		N = 740	%	N = 428	%	
Age at diagnosis	<45	167	22.57 %	165	38.55 %	<0.001
	45–54	203	27.43 %	131	30.61 %	
	55–64	219	29.59 %	80	18.69 %	
	≥65	151	20.41 %	52	12.15 %	
T stage	T1	46	6.22 %	49	11.45 %	<0.001
	T2	185	25.00 %	156	36.45 %	
	T3	185	25.00 %	111	25.93 %	
	T4	324	43.78 %	112	26.17 %	
N stage	N0	63	8.51 %	63	14.72 %	<0.001
	N1	287	38.78 %	221	51.64 %	
	N2	121	16.35 %	70	16.36 %	
	N3	269	36.35 %	74	17.29 %	
Grade	I/II	200	27.03 %	261	60.98 %	<0.001
	III/IV	540	72.97 %	167	39.02 %	
Histological type	IDC	580	78.38 %	402	93.93 %	<0.001
	ILC	46	6.22 %	2	0.47 %	
	Others	114	15.41 %	24	5.61 %	
Hormone receptor	Negative	314	42.43 %	90	21.03 %	<0.001
	Positive	426	57.57 %	338	78.97 %	
HER2 status	Negative	569	76.89 %	148	34.58 %	<0.001
	Positive	171	23.11 %	280	65.42 %	
Brain metastases	No	723	97.70 %	425	99.30 %	0.073
	Yes	17	2.30 %	3	0.70 %	
Response to chemotherapy	No response	182	24.59 %	20	4.67 %	<0.001
	Partial response	452	61.08 %	191	44.63 %	
	Complete response	106	14.32 %	217	50.70 %	

mutations) to remain relevant in rapidly evolving clinical landscapes.

The RSF model demonstrated robust predictive performance, with C-indices of 0.741 (training) and 0.720 (validation), outperforming traditional Cox regression. HER2 status, chemotherapy response, and tumor grade emerged as the top prognostic factors, consistent with their established roles in metastatic biology [21–23]. While HER2+ patients likely received anti-HER2 therapies (e.g., trastuzumab) and HR + patients endocrine therapy, SEER lacks drug-specific data. Future models should integrate treatment details to refine prognostic accuracy. Notably, the first course of radiation therapy for patients with dnMBC is palliative radiation therapy based on distant metastatic sites, with purposes that include shrinking metastatic lesions and alleviating metastasis-related events such as bone pain. However, survival benefits observed in high-risk subgroups suggest potential curative roles in select patients (advanced T/N stage, high grade, poor chemotherapy response), warranting further investigation. This finding challenges the "one-size-fits-all" approach to RT and advocates for risk-adapted therapy—a concept hitherto unexplored in dnMBC guidelines. Biomarker-driven RT, which aims at de-escalating or escalating RT, constitutes an unmet need. Although currently promising in early breast cancer [26,27], much work needs to be done in metastatic disease.

Currently, there are few shiny web pages based on RSF [28,29]. To bridge the gap between prognostic research and clinical practice, we developed a web application that dynamically integrates clinicopathological variables to generate individualized survival predictions. This tool addresses a longstanding barrier in dnMBC care—the lack of real-time, data-driven decision support—and empowers clinicians to tailor RT use based on patient-specific risk profiles. For example, a 55-year-old patient with T4N3 HR-/HER2-disease and poor chemotherapy response would be flagged as high-risk, prompting strong consideration of RT, whereas a HER2+ patient with complete chemotherapy response might forego RT without compromising outcomes.

4.1. Limitation

While our study leverages a large cohort, residual confounding from unmeasured variables cannot be excluded. SEER-based analyses have several inherent limitations. First, lack of RT Details: SEER does not capture RT dose, treatment volumes, and fractionation schedules, or temporal sequencing with systemic therapies, precluding protocol optimization. Second, surgical and systemic therapy technical parameters (e.g., axillary dissection extent, fields) and systemic therapy regimens (e.g., anti-HER2 agents) were unavailable. Third, metastatic Burden: Oligo-vs. polymetastatic status was undifferentiated, potentially confounding survival outcomes. Furthermore, prospective studies with granular treatment data are needed to validate our risk stratification model and refine multimodal strategies.

5. Conclusion

In conclusion, this study redefines the therapeutic landscape of dnMBC by establishing postoperative RT as a risk-stratified intervention and introducing the first machine learning-based prognostic tool for this population. By aligning locoregional therapy with individualized risk profiles, we advance the paradigm of precision oncology, ensuring that aggressive interventions are reserved for patients most likely to benefit. Our web application serves as a scalable solution to translate complex prognostic data into actionable clinical decisions, ultimately improving outcomes for a historically underserved patient population. As a proof-of-concept study, our risk stratification model and web tool require validation in prospective trials or large, multi-center observational cohorts.

CRediT authorship contribution statement

Chaofan Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yusheng Wang:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Biyun Fang:** Visualization, Software. **Mengjie Liu:** Visualization, Software. **Shiyu Sun:** Data curation. **Jingkun Qu:** Writing – review & editing, Supervision. **Shuqun Zhang:** Writing – review & editing, Supervision, Funding acquisition. **Chong Du:** Writing – review & editing, Validation, Investigation, Funding

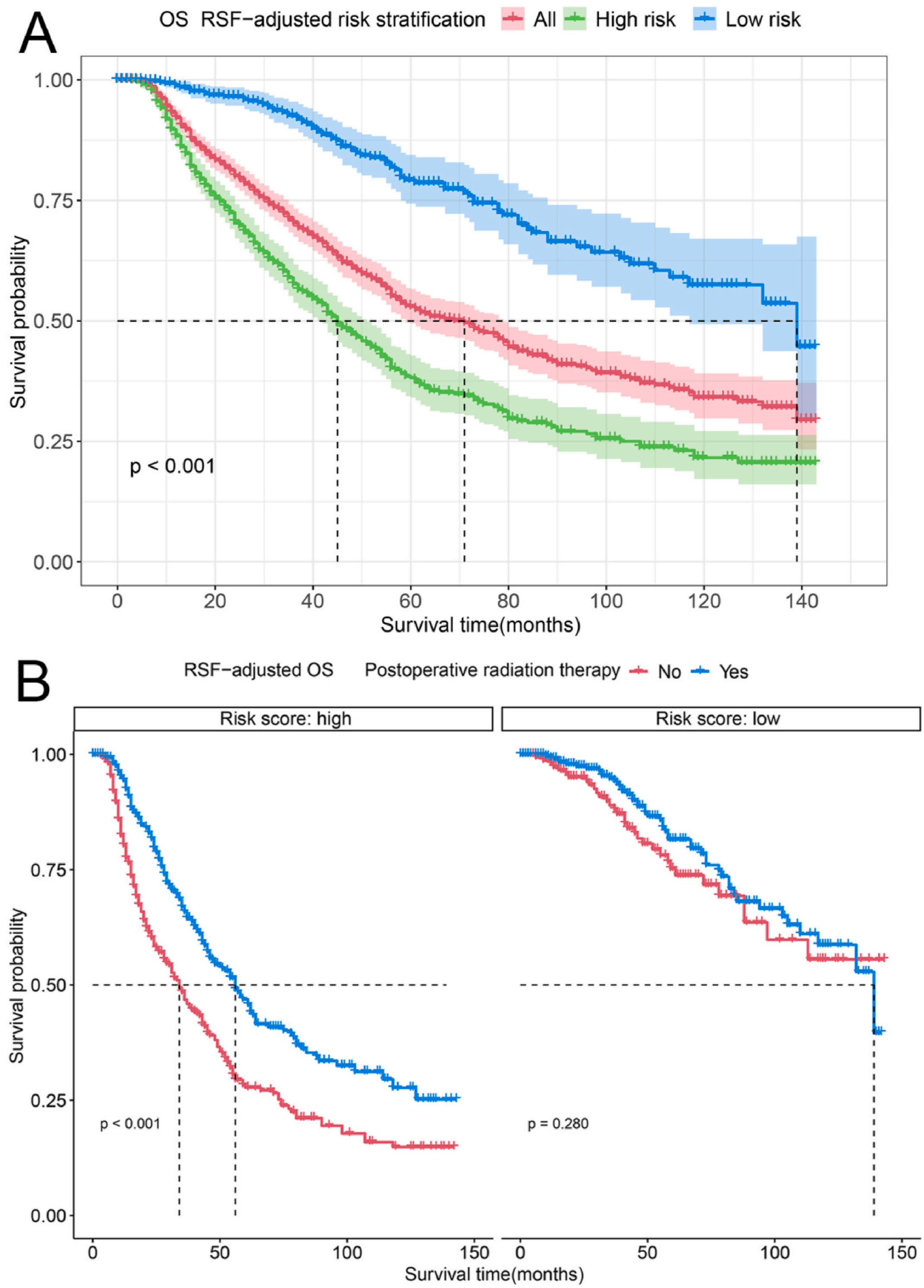

Fig. 5. Survival analysis of patients in high- and low-risk groups adjusted by RSF (radiation therapy vs. no radiation therapy)
Kaplan-Meier (K–M) survival analysis:A. RSF-adjusted OS of patients; B. RSF-adjusted OS of patients (stratified by postoperative radiation therapy); RSF: random survival forest; OS: overall survival.

acquisition.

Ethical approval and consent to participate

The data from the SEER database is publicly available. Thus, the present study was exempted from the approval of local ethics committees of the Second Affiliated Hospital of Xi'an Jiaotong University.

Consent for publication

Not applicable.

Availability of data and material

Publicly available datasets were analysed in this study. This data can

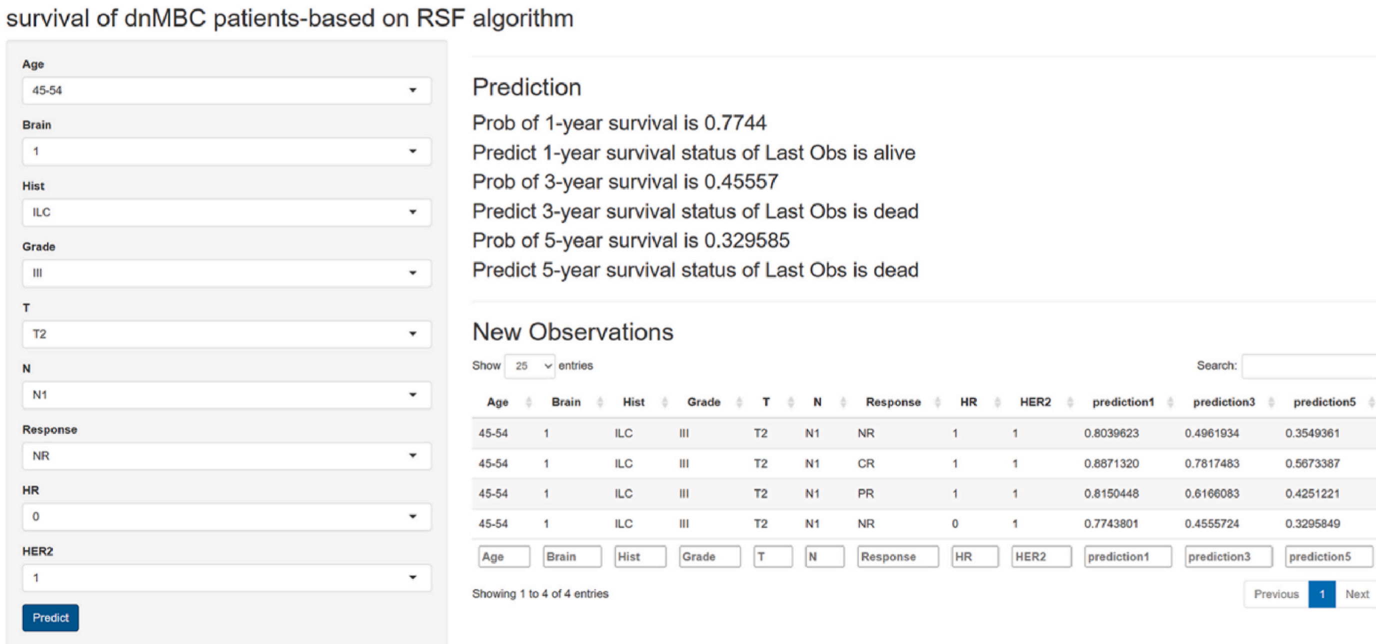

Fig. 6. Developing user-friendly Web applications to deploy RSF model using the Shiny platform (https://lee2287171854.shinyapps.io/RSFshiny/).

be found here: https://seer.cancer.gov.

Fundings

This work was funded in part by the following: National Natural Science Foundation of China (82174164, to S.Q. Zhang), the Key Research and Development Program of Shaanxi (2022SF-411, to C.D.), Natural Science Foundation of Shaanxi Province of China (2023-JC-YB-758, to B.F.), Construction project of Key Laboratory of Tumor Prevention and Treatment of Integrated Traditional Chinese and Western Medicine of Shaanxi Province (2022-ZXY-SYS-002), Shaanxi Provincial Science and Technology Innovation Team (2024RS-CXTD-89).

Declaration of Competing interests

The authors declared that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Acknowledgement

We would like to thank all the developers of the R programming package for selflessly sharing their code.

References

[1] Siegel R, Naishadham D, Jemal A. Cancer statistics. CA Cancer J Clin 2013;63(1): 11–30. 2013.
[2] Thomas A, et al. Initial surgery and survival in stage IV breast cancer in the United States, 1988-2011. JAMA Surg 2016;151(5):424–31.
[3] Malmgren JA, et al. Differential presentation and survival of de novo and recurrent metastatic breast cancer over time: 1990-2010. Breast Cancer Res Treat 2018;167 (2):579–90.
[4] Dawood S, et al. Survival differences among women with de novo stage IV and relapsed breast cancer. Ann Oncol 2010;21(11):2169–74.
[5] Caswell-Jin JL, et al. Change in survival in metastatic breast cancer with treatment advances: meta-analysis and systematic review. JNCI Cancer Spectr 2018;2(4): pky062.
[6] Li C, et al. Deep neural network provides personalized treatment recommendations for de novo metastatic breast cancer patients. J Cancer 2024;15(20):6668–85.
[7] Harris E, Barry M, Kell MR. Meta-analysis to determine if surgical resection of the primary tumour in the setting of stage IV breast cancer impacts on survival. Ann Surg Oncol 2013;20(9):2828–34.
[8] Xiao W, et al. Primary tumor resection in stage IV breast cancer: a systematic review and meta-analysis. Eur J Surg Oncol 2018;44(10):1504–12.
[9] Reinhorn D, et al. Locoregional therapy in de novo metastatic breast cancer: Systemic review and meta-analysis. Breast 2021;58:173–81.
[10] Gera R, et al. Locoregional therapy of the primary tumour in de novo stage IV breast cancer in 216 066 patients: A meta-analysis. Sci Rep 2020;10(1):2952.
[11] Cardoso F, et al. 4th ESO-ESMO international consensus guidelines for advanced breast cancer (ABC 4). Ann Oncol 2018;29(8):1634–57.
[12] Choi SH, et al. Locoregional treatment of the primary tumor in patients with de novo stage IV breast cancer: a radiation oncologist's perspective. Clin Breast Cancer 2018;18(2):e167–78.
[13] Gnerlich J, et al. Surgical removal of the primary tumor increases overall survival in patients with metastatic breast cancer: analysis of the 1988-2003 SEER data. Ann Surg Oncol 2007;14(8):2187–94.
[14] Li C, et al. Novel models by machine learning to predict prognosis of breast cancer brain metastases. J Transl Med 2023;21(1):404.
[15] Li C, et al. Machine learning predicts the prognosis of breast cancer patients with initial bone metastases. Front Public Health 2022;10:1003976.
[16] Criscitiello C, et al. Surgery of the primary tumour in de novo metastatic breast cancer: To do or not to do? Eur J Surg Oncol 2015;41(10):1288–92.
[17] Blanchard DK, et al. Association of surgery with improved survival in stage IV breast cancer patients. Ann Surg 2008;247(5):732–8.
[18] King TA, et al. Prognostic impact of 21-gene recurrence score in patients with stage IV breast cancer: tbcrc 013. J Clin Oncol 2016;34(20):2359–65.
[19] Lin C, et al. Subdivision of M1 stage for de novo metastatic breast cancer to better predict prognosis and response to primary tumor surgery. J Natl Compr Canc Netw 2019;17(12):1521–8.
[20] Seltzer S, Corrigan M, O'Reilly S. The clinicomolecular landscape of de novo versus relapsed stage IV metastatic breast cancer. Exp Mol Pathol 2020;114:104404.
[21] Bang YJ, et al. Trastuzumab in combination with chemotherapy versus chemotherapy alone for treatment of HER2-positive advanced gastric or gastro-oesophageal junction cancer (ToGA): a phase 3, open-label, randomised controlled trial. Lancet 2010;376(9742):687–97.
[22] Baselga J, et al. Pertuzumab plus trastuzumab plus docetaxel for metastatic breast cancer. N Engl J Med 2012;366(2):109–19.
[23] Strickler JH, et al. Tucatinib plus trastuzumab for chemotherapy-refractory, HER2-positive, RAS wild-type unresectable or metastatic colorectal cancer (MOUNTAINEER): a multicentre, open-label, phase 2 study. Lancet Oncol 2023;24 (5):496–508.
[24] Lloyd MR, et al. Precision therapeutics and emerging strategies for HR-positive metastatic breast cancer. Nat Rev Clin Oncol 2024;21(10):743–61.
[25] Morganti S, et al. PARP inhibitors for breast cancer treatment: a review. JAMA Oncol 2024;10(5):658–70.
[26] Meattini I, et al. Biomarker-directed radiotherapy in breast cancer: a narrative review. JAMA Oncol 2025;11(3):329–39.

[27] Aristei C, et al. Personalization in modern radiation oncology: methods, results and pitfalls. Personalized interventions and breast cancer. Front Oncol 2021;11: 616042.

[28] Chen Z, et al. Utilizing explainable machine learning for progression-free survival prediction in high-grade serous ovarian cancer: insights from a prospective cohort study. Int J Surg 2025.

[29] Ning C, et al. Development and validation of an explainable machine learning model for mortality prediction among patients with infected pancreatic necrosis. EClinicalMedicine 2025;80:103074.